Phillip H. Kieval

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Research

AOS Philosophy of AI · Philosophy of Science · Cognitive Science

AOC AI Ethics · Politics of Science and Technology · American Pragmatism

Employment

2026–present **Assistant Professor**, *Department of Philosophy, University of Guelph*

2025–2026 **Postdoctoral Associate**, *Philosophy and Ethics of AI, University of Florida*

Education

2021–2026 **PhD, History and Philosophy of Science**, *University of Cambridge*
Advisors: Marta Halina and Hasok Chang
Examiners: Rune Nyrup and Mauricio Suárez

2019–2021 **MA, Philosophy**, *University of Houston*

2015–2019 **BA, Philosophy**, *University of Houston*
Magna cum laude, with University Honors and Honors in Major

Publications

Journal articles

forthcoming "Deep Neural Networks, Architectural Constraints, and Scientific Representation." *Philosophy of Science*.

forthcoming "Formal Analogies and Model Transfer in Scientific Deep Learning." *Philosophy of Science*.

2026 "Neural Scaling Laws and Science without Theory." *Erkenntnis*. DOI

2024 "Artificial Achievements." *Analysis* 84 (1): 32–41. DOI

2022 "Permission to Believe Is Not Permission to Believe at Will." *Synthese* 200 (5): art. 347. DOI

2022 "Mapping Representational Mechanisms with Deep Neural Networks." *Synthese* 200 (3): art. 196. DOI

Book chapters

2025 "Representation Learning without Representationalism: A Non-Representational Account of Deep Learning Models in Scientific Practice." In *Philosophy of Science for Machine Learning: Core Issues and New Perspectives*, edited by Juan M. Durán and Giorgia Pozzi. Synthese Library. Cham: Springer. DOI

Commentaries

2025 With Cameron Buckner. "'Captured' by Centaur: Opaque Predictions or Process Insights?" *Journal of Experimental Psychology: Animal Learning and Cognition*. DOI

Awards and fellowships

2025 **AI Fellow**, London AI and Humanity Project, University of London (declined)

2022–2025 **Student Fellow**, Leverhulme Centre for the Future of Intelligence, University of Cambridge

2022 **Best Student Paper Award**, Issues in XAI #4, TU Delft

2021–2025 **Gates Cambridge Scholarship** (full-cost doctoral scholarship), University of Cambridge

2018–2019 **Mellon Research Scholars Fellowship**, Andrew W. Mellon Foundation

Presentations

Refereed

Nov 2026 "Deep Neural Networks, Architectural Constraints, and Scientific Representation." Philosophy of Science Association Biennial Meeting, San Diego, CA

Jul 2026 "Rationalization in Reasoning LLMs." Workshop on Philosophy of Explainable AI: New Directions, Aarhus University, Aarhus, DK

Feb 2026 "What Is Dynamical Computation?" Southern Society for Philosophy and Psychology, Atlanta, GA

Nov 2024 "Deep Learning as Method-Learning." Philosophy of Science Association Biennial Meeting, New Orleans, LA

Mar 2024 "Data Modelling and Exploratory Concept Formation in Neuroscience." Southern Society for Philosophy and Psychology

2022–2023 "Getting Real about Neural Data Science: Neural Manifolds and Exploratory Concept Formation in Cognitive Neuroscience."
- International Society for the Philosophy of the Sciences of the Mind, 1st Web Conference (Nov 2023)
- 7th International Conference on the History and Philosophy of Computing, Warsaw University of Technology (Oct 2023)
- 37th Boulder Conference on the History and Philosophy of Science, University of Colorado Boulder (Sep 2022)

May 2023 "Varieties of Representational Risk in Medical AI." PhilInBioMed 4, University of Cambridge

Nov 2022 "Automated Exploration, Understanding, and Semantic Opacity." Philosophy of Science Association Biennial Meeting, Pittsburgh, PA

2022 "Biased by Design: Why Algorithms Are Necessarily Value-Laden."
- Graduate Workshop on Philosophy of Machine Learning, Max Planck Institute for Intelligent Systems, University of Tübingen (Oct 2022)
- Ethical Issues in AI and Computing, Embedded EthiCS @ Harvard and MIT SERC, Harvard University (Jun 2022)

2021–2022 "Unsupervised Discoveries, Understanding, and Semantic Opacity."
- Issues in XAI #4: Explanatory AI between Ethics and Epistemology, TU Delft (May 2022)
- Rotman Graduate Conference: Models and Idealization, Western University (May 2022)
- 4th Scientific Understanding and Representation (SURe) Workshop, Fordham University (Apr 2022)
- Philosophy in Informatics VI, Pontifical University of John Paul II in Kraków (Dec 2021)

Oct 2021 "Artificial Achievements." Rice–UH Graduate Conference, Rice University

Invited

May 2023 "Towards Trustworthy Machine Learning: Reliability and the Value of Uncertainty." Gates Day of Research, University of Cambridge

Jun 2022 "Getting Real about Neural Data Science." Kinds of Intelligence Retreat, Leverhulme Centre for the Future of Intelligence, University of Cambridge

May 2022 "Biased by Design: Why Algorithms Are Necessarily Value-Laden." Gates Day of Research, University of Cambridge

Feb 2022 "Unsupervised Discoveries and Semantic Opacity in Data-Driven Neuroscience." Poldrack Lab, Department of Psychology, Stanford University

Teaching

University of Guelph (instructor)

Winter 2027 Language Models and Cognitive Science; Ethics of Artificial Intelligence

Fall 2026 Critical Thinking; Digital Wisdom: How to Use AI Critically and Responsibly

University of Florida (instructor)

Spring 2026 Ethics, Data, and Technology

University of Michigan (course assistant)

Winter 2025 Generative AI and Cognitive Science (with Chandra Sripada and Richard Lewis)

University of Cambridge (supervisor)

2022–2023 Introduction to History and Philosophy of Science (Fall 2022, Winter 2023, Spring 2023)

Winter 2022 Ethics and Politics of Technology

University of Houston (graduate teaching assistant)

2019–2021 Logic I (Fall 2019, Spring 2020, Spring 2021); Introduction to Philosophy (Fall 2020); grading for Introduction to Philosophy and Philosophy and the Arts (Summer 2020)

Service

Department and university

2026–2027 **Chair**, Graduate Placement Committee, Department of Philosophy, University of Guelph

2026–2027 **Member**, Inclusion and Equity Committee, Department of Philosophy, University of Guelph

2023–2024 **PhD Student Representative**, Department of HPS, University of Cambridge

2022–2024 **Organizer**, HPS Workshop; **Co-Organizer**, Jobseekers' Coffee, Department of HPS, University of Cambridge

2022 **Member**, Wellness, Inclusion, Diversity, and Equity Committee, Leverhulme Centre for the Future of Intelligence

2021–2022 **Co-Director**, Learning for Purpose Programme, Gates Cambridge Scholars' Council

Workshops organized

2023–2024 6th Scientific Understanding and Representation (SURe) Workshop, LSE and University of Cambridge

2023 Lingnan–Cambridge Workshop on AI in Science, University of Cambridge

Diversity and inclusion

2021–2022 **Organizing Committee**, Minorities and Philosophy (MAP), Department of HPS, University of Cambridge

2021 **Co-Organizer**, BIPOC Philosophical Film Series, UH MAP (funded by the Media and the Moving Image Initiative)

2020 **Co-Founder**, University of Houston chapter of Minorities and Philosophy

2018 **Participant**, COMPASS Undergraduate Philosophy Workshop, University of Texas at Austin

Refereeing

AI & Society, *Analysis*, *British Journal for the Philosophy of Science*, *Cognitive Systems Research*, *Erkenntnis*, *Ethics and Information Technology*, *Philosophical Psychology*, *Philosophy of Science*, *Synthese*, *WIREs Cognitive Science*

Languages and skills

Languages English (native), Spanish (B2)

Programming Python

References

Marta Halina
Professor of Philosophy of Science
Department of HPS
University of Cambridge
mh801@cam.ac.uk

Hasok Chang
Hans Rausing Professor of History and Philosophy of Science
University of Cambridge
hc372@cam.ac.uk

Cameron Buckner
Professor and Donald F. Cronin Endowed Chair in the Humanities
University of Florida
cameron.buckner@ufl.edu